Week 5 – Random Testing

For this document, you will need to fill out the information below. Ensure you have 5 triggering numbers for each bug or you will receive zero points for the bug. Your theory must fit the 5 provided numbers to receive any points. To receive full points, your theory must match the actual coded error, so you may need more than 5 data points for each bug to successfully determine the causes. We have provided four specific aspects of test cases you will need to correctly identify (prefix, length, checkbit status, and other).

- **Bug 1**
  - **Triggering credit card numbers (at least 5)**
    - `222091729602795`
    - `221989252604254`
    - `221861480550057`
    - `221708713908327`
    - `221687536249238`
  - **Theory that explains what triggered the bug**
    - Prefix: MasterCard prefix of 22
    - Length: 15
    - Checkbit status: Invalid
    - Other: None
- **Bug 2**
  - **Triggering credit card numbers (at least 5)**
    - `4041470701890897`
    - `4048511584278701`
    - `4040446740411938`
    - `4044693437781554`
    - `4044982312517187`
  - **Theory that explains what triggered the bug**
    - Prefix: Visa Card prefix of 404
    - Length: 16
    - Checkbit status: Valid
    - Other: None
- **Bug 3**
  - **Triggering credit card numbers (at least 5)**
    - 270677087771357
    - 295584994981357
    - 8869068497763579
    - 1412228130823579
    - 9531348627953579
  - **Theory that explains what triggered the bug**
    - Prefix: Random/Any
    - Length: 15
    - Checkbit status: Invalid
    - Other: None

- **Bug 4**
  - **Triggering credit card numbers (at least 5)**
    - `182778377315818`
    - `782148162745078`
    - `734656902651073`
    - `134885542113413`
    - `526971587564452`
    - `681775901751468`
  - **Theory that explains what triggered the bug**
    - Prefix: Any and Random
    - Length: 15
    - Checkbit status: Valid and Invalid
    - Other: None
- **Bug 5**
  - **Triggering credit card numbers (at least 5)**
    - `5131955806865727`
    - `4128657030826297`
    - `4386571536753607`
    - `4865794932411220`
    - `340865728910436`
  - **Theory that explains what triggered the bug**
    - Prefix: prefix of Visa, MasterCard, or American Express
    - Length: 15 or 16
    - Checkbit status: Valid
    - Other: None
- **Bug 6**
  - **Triggering credit card numbers (at least 5)**
    - `2275365546251258`
    - `2332132239787258`
    - `346825507255258`
    - `4598597221782258`
    - `4744941010390258`
  - **Theory that explains what triggered the bug**
    - Prefix: Any/Random
    - Length: 15 or 16
    - Checkbit status: Valid
    - Other: None
- **Bug 7**
  - **Triggering credit card numbers (at least 5)**
    - `263992315008797`
    - `529308770927886`
    - `407788945435534`
    - `557660703775138`
    - `265067303391766`
  - **Theory that explains what triggered the bug**

        - Prefix: Any/Random
        - Length: 15
        - Checkbit status: Valid
        - Other: None
- **Bug 8**
    - **Triggering credit card numbers (at least 5)**
        - `6761884774836173`
        - `9054157667576600`
        - `5252291001993914`
        - `1975653443585516`
        - `6386108008050951`
    - **Theory that explains what triggered the bug**
        - Prefix: Any/Random
        - Length: 16
        - Checkbit status: Invalid
        - Other: None